



Fortify Your Network Defence with Invisiron's Cyber Defence Platforms

INVISIRON WHITE PAPER
version 1

INTRODUCTION

Many of the security solutions on the market today focus on threat detection and do not provide defence mechanisms to automatically mitigate cyber attacks and to keep attackers out of the network. Instead, most legacy solutions focus on passive threats detection and rely on the use of a Security Information and Event Manager (SIEM) software suites, coupled with manpower, to analyse and correlate these threats. This is insufficient and ineffective at protecting against any of the more sophisticated attack types we see today.

Invisiron's Cyber Defence platforms have been designed using state of the art cybersecurity techniques. This paper will outline and describe how Invisiron protects a network from the more sophisticated cyber-attacks we see today. It will identify some of the shortfalls inherent in legacy security solutions and provide insights into the weaknesses in networks and how they can be better protected.

MODERN DEFENCE STRATEGIES

Invisiron believes that, as a primary line of defence, an organisation's network must be defended from a point as close to their Internet gateway as possible. This means installing a capable cyber defence platform immediately after the Internet router and in front of the organisation's primary switch or firewall. It may also involve installing devices strategically located immediately in front of critical assets or at key gateway points in the network. This deployment allows for preventing intrusions and mitigating attacks before they reach inside the network or asset. Stopping attacks before they take a foothold in the network is the foundation of modern cyber defence. Most people will agree that keeping an intruder out of the network should be the primary goal of any security solution.

Invisiron's Cyber Defence platforms have been designed in a unique way that allows them to be installed inline between an Internet router and a firewall, in front of critical assets, or at key gateway points in the network. Packets can be inspected in real-time as they pass through the platform and in both directions (inbound and outbound protection). Malicious packets can be dropped instantly before they have a chance to enter the network or infect a key asset and before vital information leaves the network.

It is crucial to monitor packets flowing in both directions at your organisation's Internet gateway. Often it is believed that network security is about monitoring incoming packets from the Internet. This is only half the truth. It is equally important to monitor outgoing traffic as well. For example, should a malware successfully infiltrate the network i.e. via a thumb drive or other media, there is then a chance it can be detected by monitoring the outgoing traffic it generates.

The current state of the art in defence strategies require in-line hardware-based protection which can not only analyse all traffic in both directions but can do so at or near line rate (no impact on system latency). The system must, at a minimum, provide detection engines for malicious data signatures, DGA generated domains, tracking and blocking of Tor exit nodes, targeted rate limiting, as well as traditional IP & Domain reputation detection. More advanced methods should include the ability to expand the detection of packet signatures to more complex rules to provide detection and protection against protocol-based attacks or pre-indicators of a potential attack such as scan attempts. Importantly all Invisiron's Cyber Defence platforms provide these functionalities and more while being capable of monitoring and protecting traffic in both directions (bi-directionally) at any point in the network.

LEGACY DEFENCE PRODUCTS

Before going into more details on how Invisiron's Cyber Defence solution can protect your systems, this paper will provide an outline and comparison with legacy security technologies which many organisations still utilise today.

SOFTWARE ENDPOINT PROTECTION

Software-based security technologies provide limited protection, in that many devices in an organisation's information technology ecosystem cannot be protected by software alone. Examples are network-attached storage, printers, webcams and other Internet-Of-Things (IoT) devices. Additionally, software protection solutions are often rendered ineffective by more sophisticated hacks before the attacks or new threats find ways to exploit vulnerabilities in lower system layers which the software has no control over. An ever more common attack technique is to penetrate a network using an IoT device, such as a network-attached printer, and use it as a base for further penetrations, lateral movement inside the network, and even data exfiltration.

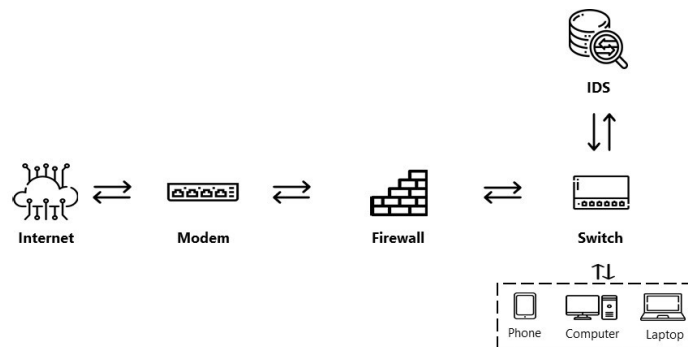
In other examples, we have seen Internet-connected webcams being integrated with extremely poor security awareness. Many of these devices use fixed factory set login credentials or password reset backdoors. On a global scale, these devices are regularly being used in distributed denial of service (DDoS) attacks, crypto mining hijacking, as well as new emerging threats created by the permeation of IoT solutions throughout the world's IT systems. Software layer protection is not capable to stop a sophisticated attacker from infiltrating and taking control of such devices. Nor is it effective against attacks such as an attack on unpatched OS vulnerabilities, firmware bugs or backdoors etc. The bottom line is, a network must be defended immediately at the Internet connection, as well as at critical points, to stand a chance of blocking attacks.

LOG COLLECTION & SECURITY INFORMATION EVENT MANAGEMENT (SIEM)

Another popular approach is the use of log collection and SIEM software tools. The premise is to collect log information from as many devices as possible and analyse this data to detect malicious activities in the network. This offers no protection against penetrations and most other types of attacks. Without an inline device that can analyse and drop or block malicious packets, protection is not possible. Most of these systems focus on providing visibility into what **has** taken place in the network in the past and requiring costly and manpower intensive manual responses. The response delays alone in this approach renders most attacks over long before they can be stopped or mitigated.

INTRUSION DETECTION SYSTEMS

Many older legacy solutions focus on providing intrusion detection only, without any mitigation. They are typically marketed as Intrusion Detection Systems (IDS) or Network Intrusion Detection Systems (NIDS) and are still being deployed in many networks today. These devices are normally installed via a tap port on a switch. A tap port is a special port on a switch that can be configured so that it sends out mirrored copies of most network traffic. A tap port is a special port on a switch that can be configured so that it sends out mirrored copies of most network traffic.



Typical Network Monitoring Installation from a Local Area Network (LAN) Tap

There is limited or no value in knowing that the network has already been breached. Invisiron's approach is to instead disrupt the attacks before they have a chance to enter the network. Most IDS/NIDS systems on the market today do not protect a network, as they are not capable of blocking malicious network traffic. Instead, they aim to detect an attack, determining that a breach has occurred. The expected mitigation then requires human intervention to act accordingly. Unless someone is monitoring these systems around the clock (24/7), it might be hours or even days or weeks before a network administrator is made aware that their network has been breached. Additionally, a serious drawback of these systems is that as a rule, tap ports on switches have a lower priority to do packet mirroring so as not to cause network disruption. As a result, in modern networks with their ever more increasing data rates, many of the packets which pass through the switch are not mirrored and as such never seen by the IDS/NIDS or

connected SIEM system. Passively listening to network traffic alone is inadequate in countering today's threat landscape.

FILE SIGNATURE BASED MALWARE PROTECTION

Another category of security technology is software utilities which are installed directly on the host machines which base their protection on file signatures. If, for example, a user downloads a file from the Internet, a checksum (signature or fingerprint) of the file will be calculated by the software. This signature is then sent to a centralised server owned by the company that provides the security software. It will be compared against other signatures from both known good files and against signatures of known malicious files. This type of protection does deliver some value (albeit limited), provided that the vendor's database is updated fast enough with new threat signatures and contains a large enough database of signatures from the past. It is however very easy to bypass this defence. All the hacker has to do is slightly rearrange their malicious file so that it generates a different file signature, which will then easily bypass any signature-based protection solution.

Additionally, this type of signature-based protection is not capable of detecting attacks that are not delivered through a file. It is also not capable of detecting threats that do not use persistence, meaning that they do not store information on the system's hard disk drive. Furthermore, network penetration or breaches cannot be detected by these solutions.

Another category of signature-based defence products sends complete files from the owner's computer up to a server for offline pattern matching. There have been many reports in the news media where high profile companies have had important payroll files and other confidential data exfiltrated out of their network and sent to the security software vendor's signature servers for analysis. Such method is not viable in today's modern privacy-conscious landscape. Furthermore, while this not only counter too many privacy regulations hurdle, in some cases this approach can actually increase the risk of certain types of attacks. If the network contains sensitive data and files, one must be aware that this data might be exfiltrated out to be analysed externally by some systems.

In summary, signature-based malware protection has little value when used alone. It should rather complement properly placed hardware systems and be part of an overall defence strategy.

MODERN CYBER SECURITY SOLUTIONS

In today's sophisticated attack environment, protecting an organisation's networks is no simple task. The first and most important step is endeavoring to keep attackers out of the organisation's networks. It is of the highest importance that Cyber Defence mechanisms include inline devices positioned as close to the Internet gateway as possible to allow them to detect and prevent attacks before they reach inside a network. It is also of very high importance that these platforms are capable of automatically blocking malicious traffic by stopping the network packets as soon as they are detected as being malicious. If such packets can enter a network, the damage may already be done. These packets must be stopped and dropped as soon as possible before they have a chance to infiltrate the network.

Some inline installed defence systems suffer from performance degradation, and/or high failure risks. These weaknesses have been overcome by Invisiron's unique design architecture, further details on these important points can be found in the section "**Inline Performance**" of this paper.

MINIMUM NETWORK DEFENCE CAPABILITIES

A proper defence implementation for any network must include, at minimum, seven different types of advanced cybersecurity capabilities:

1. A traditional malicious IP address tracking engine
2. A malicious Domain tracking engine
3. A malicious URL tracking engine
4. A domain generating algorithm (DGA) detection engine
5. A Tor traffic detection and filtering engine
6. A counterfeit SSL certificate tracking engine
7. A deep packet inspection (DPI) engine

These engines all complement each other and offer very strong protection.

One of the common complaints by security auditors or researchers after an attack has occurred is the lack of evidence available to determine exactly what has transpired and improve future defences. Invisiron's Cyber Defence platforms include full packet logging capabilities where the entire network traffic for an attack is kept in a standard PCAP format which can be interpreted by network analysis tools. This is vital in improving future protections and determining exactly what has transpired particularly in cases of more sophisticated attacks.

Cyber threat intelligence (CTI) is another key part in a comprehensive Cyber Defence. CTI is the act of keeping the aforementioned engines constantly updated with new threat types. More details on these engines and systems follow.

REPUTATION DETECTION ENGINE

The reputation detection engines provide a layer of protection by validating the packet source and destination IP addresses, domain names and URLs against lists of known malicious instances. Invisiron's Cyber Defence platforms contain a database of known malicious domains, URLs, and IP addresses. The database typically includes tens of thousands of IP addresses and tens of thousands of malicious domains and URLs.

This database is provided to Invisiron's Cyber Defence platforms through Invisiron's CTI feed located in the cloud. The cloud server harvests real-time threat information from various sources. This information is then vetted and converted into a unified format that is suitable for sending to the security platforms. Each active platform will poll the cloud server and within the hour update its database with the latest reputation lists to be protected against new and emerging threats.

It is important to note that the reputation engine is only as good as the intelligence that informs this engine about the reputation of domains, IP's, and URL's. Invisiron's reputation engine is informed by high-quality cyber indications and warning data from validated and trusted independent sources (i.e. research firms, associations, trusted agents, government contacts, etc.), together with Invisiron's own home grown CTI. This data is fully reviewed, validated, and integrated into the platforms.

The reputation detection engine informed by Invisiron's cyber threat intelligence feed checks incoming and outgoing network packets against this database. If a hit is detected, meaning a packet is very likely to be malicious and unwanted, the engine will cause this packet to be dropped. It is of prime importance that these malicious packets be dropped immediately to ensure that they will not cause any harm in the network.

DOMAIN GENERATING ALGORITHM (DGA) DETECTION ENGINE

The Domain Generating Algorithm, or DGA, engine is responsible for detecting domain names that are algorithmically generated by a special DGA software. These DGA software are typically found in more complex malwares such as ransomwares and any malware designed to call home to a command and control server (C2 server) on the Internet.

A DGA software generates very unusual looking domain names which the malware then uses to try and connect with its C2 server. Examples of such domain names are **12k3256hv45g435g.com**, **jhjh76gjrgfhfshfh.org** and **h876tgshdutidu6t.net** which look random and unusual but they are perfectly valid domain names. The reason the domain names are created in such a way is to confuse and avoid detection. In reality, these domain names are actually not random strings - they are carefully crafted domain names where an infected computer's system time is used as a seed when generating these strings.

It is very difficult to detect these unusual looking domain names. Many attempts have been done to, for example, use statistical analysis to calculate domain name strings entropy. However, these techniques have had very little success since they have a very high false positive rate. Invisiron uses another sophisticated way of detecting DGA domains. Since DGA software engine source codes are readily available on the Internet, using these software in a controlled environment to generate its domain names ahead of time is a very good technique for detecting domains used in the wild. This is what Invisiron does. As a result, we are able to track hundreds of thousands of active DGA domains in real-time. This is therefore probably the most important and powerful features of the Invisiron's Cyber Defence platforms. Being able to detect DGA activities is priceless and have saved many of our clients from severe malware infections such as ransomwares.

TOR TRAFFIC DETECTION ENGINE

The Tor traffic engine is responsible for detecting traffic originating from within the Tor network (Tor relays). Tor is what allows users to stay anonymous when using the Internet. Tor enables technologies such as the darkweb where people can acquire drugs, weapons and similar prohibited products. Hackers have also discovered that it is possible to attack a network from within Tor and therefore make it impossible to track down the culprit behind the attack.

Invisiron tracks what is called Tor endpoints. These are servers on the Internet that function as gateways between the Tor network and the normal visible Internet. In most cases there is no legit use case for allowing traffic from within Tor to reach a network. Therefore, Invisiron blocks this traffic so that it cannot reach into the protected network.

There are valid use cases for Tor. Journalists in countries suppressing free speech use Tor when they investigate their stories. Therefore, Invisiron's security platforms allow for disabling this engine if the user requests so.

SSL CERTIFICATE TRACKING ENGINE

The SSL certificate tracking and validation engine performs checks of all SSL certificates used in a network. Its purpose is to identify counterfeit certificates. Although SSL certificates are supposed to be issued by trusted well known authorities, there have been cases where some of these authorities have sold counterfeit certificates in return for monetary compensation.

This is a very serious issue and also very difficult to detect. The Invisiron's Cyber Defence platform validates each SSL certificate it sees on the Internet against an updated repository of known malicious certificates.

DEEP PACKET INSPECTION (DPI) ENGINE

The DPI engine is responsible for performing pattern detection on every network packet. The engine is rule-driven which means it is controlled by a set of externally provided inspection rules. These rules contain instructions for what to look for and where to look inside a packet.

Rules can be complex to write and understand, especially if they need to be written in a low-level syntax. Writing low-level rules is difficult and requires an in-depth understanding of the rule language syntax. To counter this complexity, Invisiron's Cyber Defence platforms incorporate an easy to use Graphical User Interface (GUI) based tool that allows for rules to be written in a much simpler way. Invisiron includes a system rule set in the cyber threat intelligence (CTI) feed. The system rule set includes a wide range of rules that protect against common attacks and penetration attempts as well as some of the more sophisticated attacks like protocol exploits. Examples of such rules are those which block port scans or OS fingerprinting.

The system rule set is updated with new rules as new attack types are detected. Examples of such attacks are the widely publicised 'Eternal Blue' type attacks, where highly advanced government level hacking tools were leaked and then used to attack the private sector and critical infrastructure. Invisiron provided rules for these attacks as soon as they became public and immediately distributed these new rules through the CTI feed. As a result, all Invisiron's Cyber Defence platforms were immunised and provides client networks protection within hours of the first attack being detected, all without requiring any intervention by the end-user.

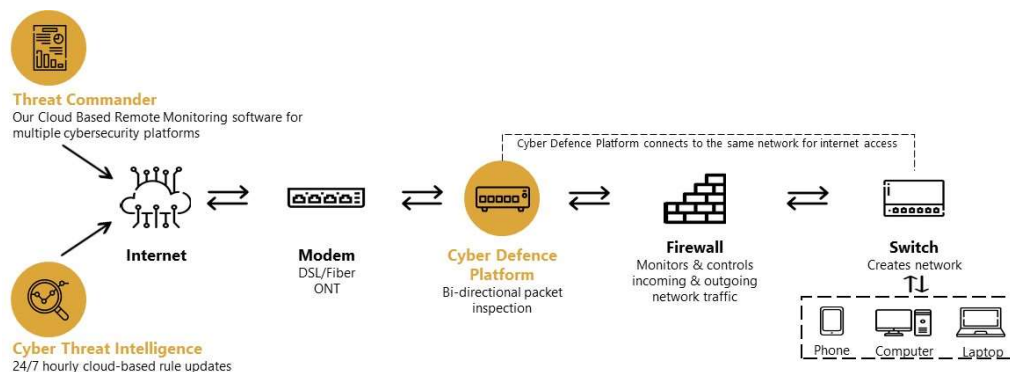
The configuration and rule writing GUI tool allow for rules to be exported out from and imported into Invisiron's Cyber Defence platforms. This means custom rules can be shared between platforms. Rule files are generated in JSON format which makes them easy to read and understand.

EVIDENCE COLLECTION AND LOGGING

This engine is responsible for collecting information relating to threats that the Cyber Defence platform detects. It generates various log files, both in text format and in PCAP format. This makes it easy to open and study the files with third party tools such as Wireshark. Log files are easily downloaded from the device. All log files are time stamped and rotated to make sure they do not grow too big and also to allow for identifying which date each log file was generated. Log files can be archived enabling the keeping of records as far back as the customer requires. Additionally, Invisiron's Cyber Defence platforms have the ability to perform a full packet capture regardless of threat such as during an advanced penetration test or network audit.

INLINE PERFORMANCE

One of the issues when installing a device inline is possible performance degradation of the Internet traffic to and from (bi-directionally) the protected network. This is the case with many legacy inline products. Invisiron's Cyber Defence platforms have been designed in a unique way to allow them to operate inline without causing any noticeable performance degradation at all. To accomplish this goal, Invisiron's platforms eliminate the use of the traditional TCP/IP network stacks in the critical path. The high-speed ethernet ports of Invisiron's Cyber Defence platforms do not require an IP address to be assigned or a Media Access Control (MAC) address to be exposed. Instead, network packets flow through the platform in stealth mode. The defence platform performs its advanced security operations directly on each packet inside the transmit and receive buffers of the high-speed network Interface controller card (NIC). As a result, Invisiron's Cyber Defence platforms remain completely invisible in the network and even with sophisticated detection tools such as NMAP, it is not possible to detect their presence.



Typical Invisiron Cyber Defence platform deployment is installed behind the Internet gateway

A common question asked is, how many computers can be protected in the network by one Invisiron device? It is assumed that each computer in the network consumes a certain amount of bandwidth in the Invisiron device and therefore it implies a limit to how many computers can be protected. This is not the case with Invisiron devices. One Invisiron device can protect any number of computers in the network with no limit constraint. The reason this question is often raised is because older legacy solutions such as old school next generation firewalls do have such limitations where performance decreases for each computer in the protected network. However, such limitation does not apply to more modern products such as Invisiron.

FAULT TOLERANCE

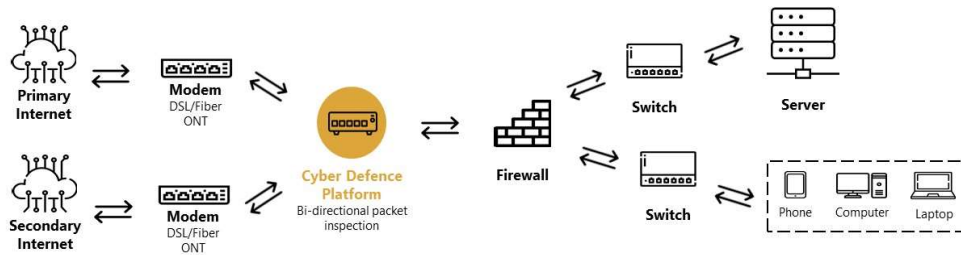
It is a common misconception that installing a security device inline and directly after the Internet modem/router increases the risk of failures and can lead to being cut off from Internet access. Another common misconception is that a security device installed inline must also include bypass circuitry that makes it possible for Internet traffic to pass through the device should it fail, or even if the power is turned off.

These assumptions are not completely accurate. It should be noted that the likelihood of being cut off from Internet access is present even without installing a security device in the critical path. The issue exists with all other key components of network operations such as the router, firewall, and core switches. Unless all these devices are duplicated in a mirrored high availability manner, there is always a risk of a single point of failure. This is not a new risk introduced by the security device, it already exists with the other components of the network.

Some Invisiron Cyber Defence platforms include high availability (HA) support. When configured in HA mode, one platform becomes the master platform and the other becomes the slave platform. The two platforms communicate with each other to determine the pair's functional status. Should the master fail, the slave immediately takes over. This happens transparently and does not require user intervention. Once the failed platform is brought back online, the system becomes the master again, provided it was the master when it failed.

REDUNDANT INTERNET CONNECTIONS

Some regions and countries struggle with less than ideal Internet services. In these situations, it might be a good compromise to add fault tolerance to the Internet connection only. This will reduce cost but still provide good protection. To support this type of fault-tolerant internet connection, two independent Internet connections can be handled by a single Invisiron platform.



Dual redundant Internet connections configurations using one S-4000

If the primary Internet connection goes down, the Invisiron platform will automatically detect this and switch over to the secondary connection. Once the primary connection resumes functionality, it will take over as the primary source for Internet traffic again.

PORT BRIDGING OR BYPASS FUNCTIONALITY

Some solutions include an automatic port bridging functionality. If the device fails or loses power, the bridging kicks in and it will still be possible to reach the Internet, no traffic will be interrupted. In modern cyber defence systems, a cyber defence platform should never be allowed to perform such bridging as this would open up the attack path towards the network and an attacker is free to penetrate the network with little effort. This is not acceptable for any network, especially larger and enterprise type networks.

Tests conducted in Invisiron's security lab, as well as by independent third parties, have demonstrated that, in the current threat landscape, if a network is left unprotected with various network ports open, it takes less than a minute before scripted attack scanners detect this. Hackers run sophisticated scripts that constantly scan the Internet for vulnerable systems. When found, an unprotected system may be immediately attacked. Therefore, it is extremely high risk to use products that allow such bypassing if they fail. The correct way to protect against device failures is to use the high availability techniques described earlier in this paper.

Sophisticated vendors such as service providers and telecom companies never allow devices that bypass the security filtering in their networks.

REMOTE MONITORING

Invisiron offers a new and modern remote monitoring software called “Threat Commander”. This software tool is designed to perform remote monitoring of multiple Invisiron’s Cyber Defence platforms in real-time. It replaces older legacy Security Information and Event Management (SIEM) style tools and focuses exclusively on real-time monitoring of security events and other events in the platforms.

Threat Commander is typically deployed on a virtual server in the cloud or on-premise in a server. Each Invisiron’s Cyber Defence platform to be monitored is configured to point towards the Threat Commander server. Once the remote monitoring is activated, the security platforms will start pushing up all its security-related data to Threat Commander.

Threat Commander is designed to manage multiple Invisiron Cyber Defence platforms and allow for drilling down into the details of each platform. Users can leverage Threat Commander to get detailed information about all security events (current and historical) recorded on the platforms. Users can also download various log files such as the system log, event logs and dropped packet logs from Threat Commander.

Threat Commander also offers several forensic or analytical functions to analyse security events in detail. This provides technical users with the ability to make informed decisions and take further actions if required. Statistics and data can be exported into comma-separated files which can be imported into other applications, for example, Excel for further processing or report generation.

Invisiron Cyber Defence platforms also support integration with various 3rd party SIEM software tools though it is strongly recommended to use Threat Commander as a targeted solution to the protection provided by our platforms.

COMPLEMENTING AN INVISIRON TECHNOLOGY BASED DEFENCE

Many potential clients ask "Is it enough to just deploy the Invisiron's cyber defence platform in my network?" The answer is no! Strong cyber defence is all about using complimentary cyber defence products that each handle a certain aspect of defence.

Invisiron Cyber Defence platforms will, at best, be able to detect and stop approximately 70-80% of attack attempts faced by organisations today. The remaining 20-30% of attacks will make their way through the defence and into the network. This may result in gaining a foothold in the network unless the defence platform is complemented by other solutions. One example is malicious emails. A user might click on an email with a specially crafted link in it. This results in a ransomware infection. The best way to protect against this is to deploy an email filtering product such as McAfee or Norton. Another example is an employee that brings in their personally owned laptop or smart phone to the organisation and connects it to the network. These devices might have been infected before it was introduced into the network. This could allow malware to enter the network and cause harm without being detected and blocked by the endpoint device. USB sticks are another source of infection, cases have been found in the past where even brand new USB sticks straight from the manufacturer have been infected by malware.

Depending on the level of security an organisation requires, complementary malware detection solutions should always be installed and deployed along with Invisiron's Cyber Defence platforms for higher degree of protection.

The logo features the word "invisiron" in a bold, lowercase sans-serif font. The letter "o" is replaced by a stylized icon of a pen nib or fountain pen tip, which is also in a bold, lowercase sans-serif font. A registered trademark symbol (®) is positioned to the upper right of the "n".

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